Given that data,

Set : V

operation : $*$

<u>closure:</u> Any $a, b \in V$ then $\underbrace{a*b}_{\text{new element}} \in V$

<u>commutativity</u> : Any $a, b \in V$ then $a*b = b*a$

<u>Associativity</u> : Any $a, b, c \in V$ then $\underbrace{(a*b)}_{\text{new element}}*c = a*\underbrace{(b*c)}_{\text{new element}}$

<u>Identity</u> : There exist a fix element e_V :

$$e_V * a = a * e_V = a$$

<u>Inverse:</u> For each element $a \in V$ there exist a unique element a' (inverse element) : $a' * a = a * a' = e_V$

$(\mathbb{R}, \times)$

identity

$e_{\mathbb{R}} = 1_{\mathbb{R}}$

dependent operation

$$(2) \times \left(\frac{1}{2}\right) = 1_{\mathbb{R}}$$

$(\mathbb{R}, +)$

$e_{\mathbb{R}} = 0_{\mathbb{R}}$

$$(2) + (-2) = 0_{\mathbb{R}}$$

Suppose, $a, b \in \mathbb{R}$, and define $\underbrace{a * b := a + b - 1}$

identity & inverse

$\boxed{e * a = a * e = a}$ $\boxed{a' * a = a * a' = e}$

Step 1

fix e,

$\underbrace{e * a} = a$

$\Rightarrow \underbrace{e + \cancel{a} - 1} = \cancel{a}$

$e = 1$

And,

$a * e = a$

$\Rightarrow \cancel{a} + e - 1 = \cancel{a}$

$\Rightarrow e = 1$

(Because both identity are same hence $e = 1$)

Step 2

Pick any $a \in \mathbb{R}$, then assume a has a inverse and denoted it by a'.

$a' * a = e$ (it's necessary to the value of e or identity)

$a' + a - 1 = 1$ (from step 1)

$\boxed{a' = 2 - a}$

find the inverse using this formula

Given

$\boxed{a * a' = e}$

check it! $a' = 2 - a$

$(2) * (0) = 1$

$2 - 2 = 0$

(Vector Space)

<u>Physicists:</u>

<u>CSE:</u> $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

<u>Math:</u> 10 Axioms $(\text{Set}, *, \odot)$

$* = +$

$a+b \in V$

$a+b = b+a$

$(a+b)+c = a+(b+c)$

0

$(-a)$

$\odot$ $\boxed{\alpha} \odot v$

fields α, β, γ

(i) closure: $\alpha \in \mathbb{F} = \{\mathbb{R}, \mathbb{C}\}$

$\alpha \odot v \in V$

(ii) $\alpha \odot (u+v) = \alpha \odot u + \alpha \odot v$

$2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$

$2\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 2 \end{pmatrix}\right) = 2\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 2 \end{pmatrix}$

(iii) $5 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$ $\quad (2+3)\begin{pmatrix} 1 \\ 2 \end{pmatrix} = 2 \cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 3\begin{pmatrix} 1 \\ 2 \end{pmatrix}$

then, $(\alpha+\beta) \odot u = \alpha \odot u + \beta \odot u$

(iv)

$$3 \odot \left(2 \odot \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right) = (3 \cdot 2) \odot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

scalar vector

vector

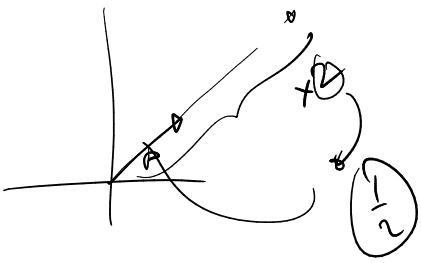
scalar

scalar scalar

$$\beta \odot (\alpha \odot u) = (\beta * \alpha) \odot u$$

(v) Identity, I_V

Any $a \in V$, then $I_V \odot a = a \odot I_V = a$



Verify Vector space

Set $V = \mathbb{R}^n$

$$\begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 y_2 \end{pmatrix}$$

$$k \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} k x_1 \\ k y_1 \end{pmatrix}$$

Q: Verify Is $(V, +, \cdot)$ a vector space or not?

(A1) is satisfy.

(A2) is satisfy.

(A3) $u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ $v = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ $w = \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$

$$(u+v)+w = \left(\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} \right) + \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 + x_2 \\ y_1 y_2 \end{pmatrix} + \begin{pmatrix} x_3 \\ y_3 \end{pmatrix}$$

$$= \begin{pmatrix} x_1 + x_2 + x_3 \\ y_1 y_2 y_3 \end{pmatrix}$$

$$u + (v+w) = \overset{\text{(Try yourself!)}}{\cdots\cdots} = \begin{pmatrix} x_1 + x_2 + x_3 \\ y_1 y_2 y_3 \end{pmatrix}$$

(A4) <u>Identity</u>: Suppose, $e_{\mathbb{R}^n} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$$e_{\mathbb{R}^n} + u = u \qquad u + e_{\mathbb{R}^n} = u$$

$$\Rightarrow \begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \alpha + x_1 \\ \beta y_1 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$$

$\alpha + x_1 = x_1$ and $\beta y_1 = y_1$

$\alpha = 0$ $\qquad$ $\beta = 1$

$$e_{\mathbb{R}^n} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(Try yourself!)

$$e_{\mathbb{R}^n} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

(A5) <u>Inverse</u>: $u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ then assume the inverse element of u is $u' = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$

$u + u' = e_{\mathbb{R}^n}$

$$u' + u = \underbrace{e_{\mathbb{R}^n}}$$

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} + \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (\text{from A4})$$

$$\begin{pmatrix} \alpha + x_1 \\ \beta y_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\alpha + x_1 = 0$ & $\beta y_1 = 1$

$\alpha = -x_1$ $\qquad$ $\beta = 1/y_1$

$$u = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \Longrightarrow u' = \begin{pmatrix} -x_1 \\ 1/y_1 \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} 2 \\ 0 \end{pmatrix}} \Longrightarrow \begin{pmatrix} -2 \\ 1/0 \end{pmatrix} \rightarrow \text{undefined}$$

For each element $u \in V$ there exist an unique element

u' such that $\quad u' * u = u * u' = e_V$

Example-2

$$\begin{pmatrix} x_1 \\ y_1 \end{pmatrix} + \begin{pmatrix} x_2 \\ y_2 \end{pmatrix} = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}$$

$$k \cdot \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} = \begin{pmatrix} kx_1 \\ y_1 \end{pmatrix}$$

$\boxed{\text{check list}}$

A3 : Associativity

A4 + A5 : Identity + Inverse

A2 : Commutativity

A1 : closure

M2: Vector Addition

M4: Compatibility

M3: Field Addition

M5: Identity

M1: Closure